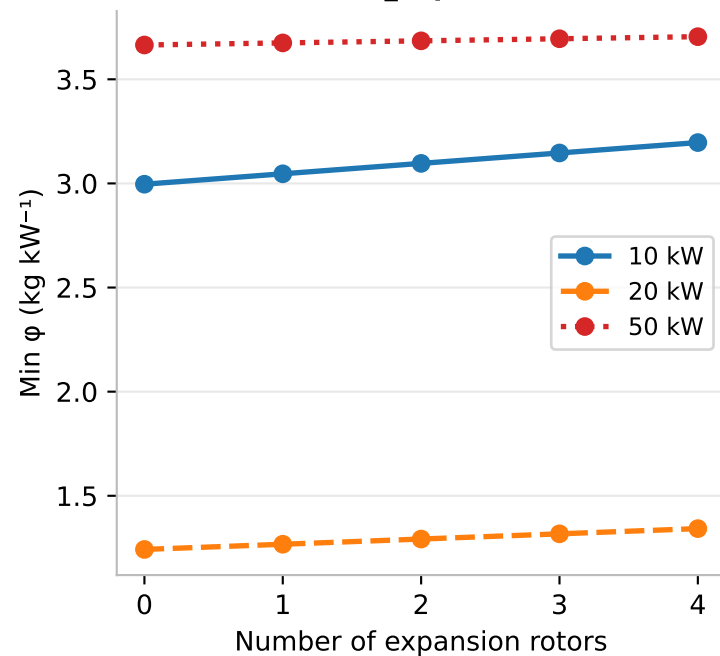


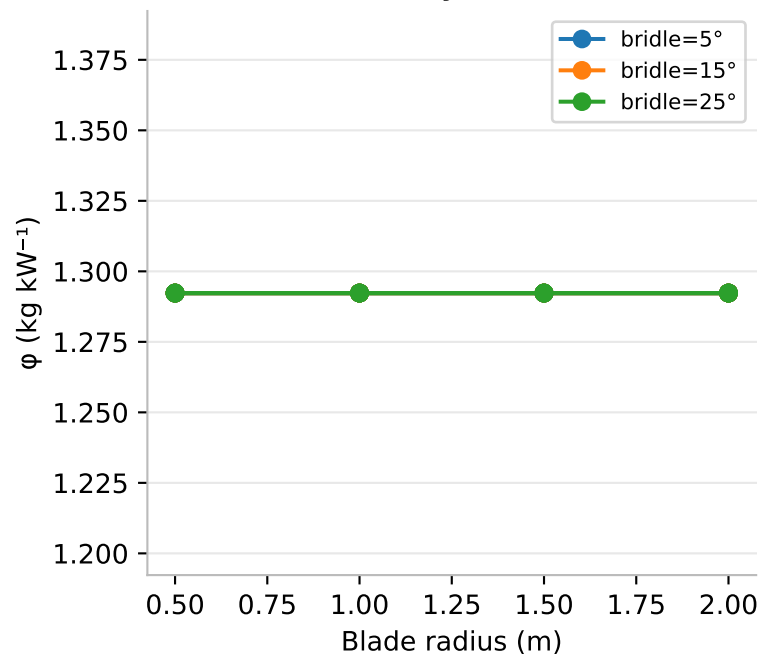
## Expansion Rotor Model Status (V6)

Current: fixed dead-weight penalty only — aerodynamic benefit not yet modelled

$\phi$  monotonically increases  
with  $n_{\text{expansion}}$



$\phi$  vs blade\_radius (20 kW,  $n_{\text{exp}}=2$ )  
Flat = no aerodynamic feedback



Expansion mass =  $n \times 0.5$  kg fixed  
(geometry-independent)

